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Influence of process parameters on the microstructure and mechanical property of Sapphire/Invar joints welded by femtosecond laser

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Metrics

Outline

Highlights

Abstract

Keywords

1. Introduction

2. Experimental procedures

3. Results and discussion

4. Mechanical property

5. Conclusions

Highlights

- High-strength bonding of sapphire/Invar was achieved using femtosecond laser welding.
- The joint is a mosaic structure of sapphire and Invar mechanically mixed without chemical reaction.
- Sapphire/Invar alloy joints with maximum shear strength up to 222.28MPa.
- The pulse duration needs to exceed the minimum welding threshold of 800fs.
- The peak power density for welding needs to be in a range from $1.49 \times 10^{12} \text{W/cm}^2$ to $5.47 \times 10^{12} \text{W/cm}^2$.

Abstract

This study achieved a high-strength sapphire/Invar alloy bonding through femtosecond laser welding. This study examined the impact of average laser power, pulse width, and pulse frequency on the macroscopic and microscopic morphology, as well as the joint shear strength of the joints, alongside the mechanisms of joint formation. The study results indicate that the joint is a mosaic structure composed of a mechanical mixture of sapphire and Invar, which does not experience chemical metallurgical reactions. The parametric study indicates that excessively short pulse durations lead to insufficient single pulse-material interaction times. Therefore, the pulse width must be adjusted to meet the minimum welding threshold of 800fs, while the peak power density should be regulated between $1.49 \times 10^{12} \text{W/cm}^2$ and $5.47 \times 10^{12} \text{W/cm}^2$ to achieve optimal joints characterized by distinct mosaic structures. When the average power is 11W, the pulse duration is 800fs, and the pulse frequency is 200kHz, the joint shear strength attains a maximum of 222.28MPa, significantly exceeding the joint shear strength of previous work. It is hoped that this study can provide a reference for selecting parameters in ultrafast laser welding of transparent materials to metals.

CRediT authorship contribution statement

Declaration of competing interest

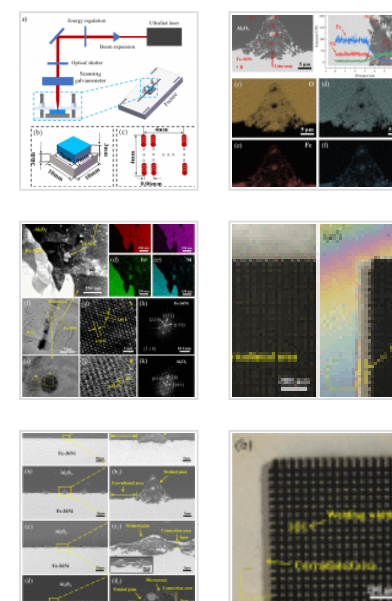
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Data availability

References

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Figures (11)



[Show 5 more figures](#) ▼

Tables (2)

Table 1. Femtosecond laser experimenta...

Table 2. EDS results at Points A-F marke...

Keywords

Femtosecond laser welding; Sapphire; Invar; Parameter optimization; Mosaic structure; High-strength bonding

[< Previous article in this issue](#)

[Next article in this issue >](#)

1. Introduction

Sapphire, a monocrystalline form of aluminum oxide, is a transparent ceramic that finds extensive application across various fields such as aerospace, medical devices, electronic product screens, missile windows, and microwave device windows, owing to its superior optical, mechanical, chemical, and thermal stability characteristics [[1], [2], [3], [4]]. On the other hand, Invar is distinguished by its minimal coefficient of thermal expansion (CTE), attributed to the unique Inka effect [[5], [6], [7]]. This property renders Invar exceptionally suitable for applications requiring dimensional stability across a range of service temperatures, including but not limited to space cameras in aerospace engineering, precision instruments, optical equipment, and metrology devices [8,9]. The joining process of sapphire and Invar alloys with high stability, is paramount in aerospace and precision instrumentation.

However, given the inherent hardness and brittleness of sapphire, its predominant joining techniques include brazing [10,11] and diffusion bonding [12,13]. The core of these methodologies is to employ an intermediary substance that facilitates the union between sapphire and the other material, circumventing the challenges posed by directly merging two substances with markedly distinct CTE. Nonetheless, both brazing and diffusion welding necessitate the application of elevated temperatures, culminating in significantly increased thermal stress within the joint. Concurrently, the interaction between sapphire and the intermediary substance engenders the formation of a novel phase lattice

inconsistent with that of sapphire, thereby exacerbating the residual stress levels within the joint. Consequently, advancements need to be undertaken to enhance connection reliability, as well as to augment processing precision and efficiency. Such improvements are essential for facilitating sapphire's effective integration and utilization in practical applications, extending its applicability and performance in relevant fields.

Ultrafast laser welding has received significant attention in recent years. This is primarily due to the unique capabilities of ultrafast lasers in achieving nonlinear energy deposition in transparent materials, as outlined in several studies [\[14\]](#), [\[15\]](#), [\[16\]](#), [\[17\]](#), [\[18\]](#), [\[19\]](#), [\[20\]](#). It offers great potential in the high-precision welding of transparent materials. This technology enables welding precision at the microscale accuracy. Additionally, it provides greater bonding strength while reducing the heat-affected zone, which contributes to the welding of transparent materials. During the joining process, the ultrafast laser traverses the transparent material, primarily interacting at the interface of the two entities. This interaction leads to the formation of a dense plasma that is essential for energy input and subsequent melting of the materials. Under the stirring of the laser beam, the molten material mechanically mixes with the plasma, and upon cooling, a welded joint is created, as documented in the literature [\[21\]](#), [\[22\]](#), [\[23\]](#). Recently, an increasing number of studies are attempting to extend the material combinations for ultrafast laser joining from homogeneous transparent materials to combinations of transparent and non-transparent materials.

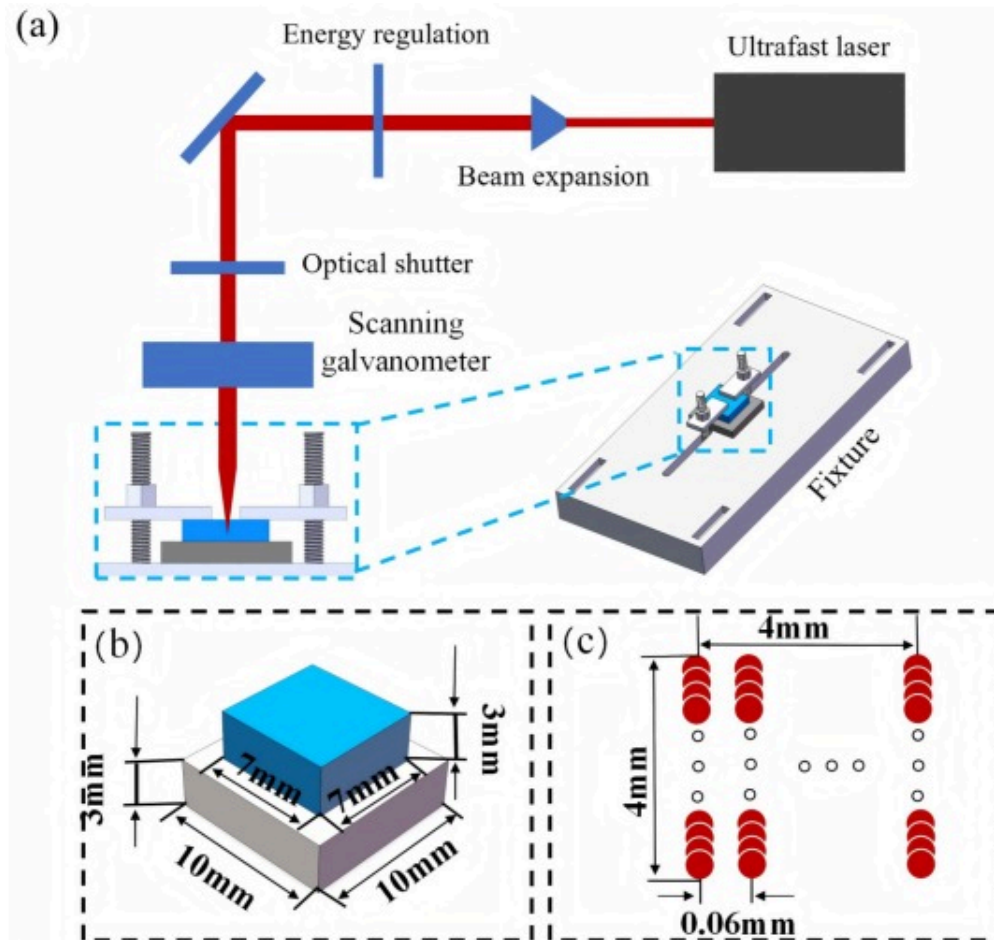
In 2016, Octav P. Ciuca et al. [\[23\]](#) used picosecond lasers to achieve a joint between glass and aluminum alloy. They showed that the formation of the weld included the formation of a confined plasma cavity at the interface of the joint, rapid heating by nonlinear absorption in the glass and the generation of multiple secondary keyholes in the metal substrate by beam scattering. In 2021, Zhang et al. [\[22\]](#) used a millisecond laser to prepare a laser-sealed joint between glass and aluminum alloy with a mosaic structure. The results showed that the laser power, frequency, and pulse duration had an essential effect on obtaining the highest bond strength. Successful sealing of high alumina ultra-thin glass with 6061 aluminum alloy was achieved. In 2023, our team [\[21\]](#) used femtosecond

pulsed laser to directly bond sapphire/Invar alloy with a serrated interface and a maximum shear strength of 108MPa. In the same year, a nanosecond laser was used to achieve a non-optical contact direct connection between glass and copper, forming a mechanically interlocked structure through a spindle-shaped weld with a maximum shear strength of 33.4MPa, which is 81.26% of the glass substrate [24]. In 2023, Zhang et al. [25] used femtosecond lasers to achieve a connection between quartz glass and an aluminum alloy, which resulted in a significant diffusion and melting pool zone and improved the strength of the joint. In 2024, Zhang et al. [26] used a femtosecond laser to realize the joining of aluminum alloy and molten silicon, revealing the joint structure. The joint was divided into three bonding regions, where the primary connection mechanism of the joint was element diffusion, characterized by many Al-Si-O transition zones.

However, it can be seen that most research focuses on the laser micro-joining of glass-to-glass and glass-to-aluminum alloy materials, with limited studies addressing the joining of transparent ceramics to metals. Previous work [21] by our research group on sapphire/Invar alloy joints just provided a basic understanding of the bonding interface but did not give an insight in the evolution of the joint with laser parameters. Consequently, this study aims to deepen the understanding of femtosecond laser welding of sapphire to Invar alloy by methodically varying the laser's power, frequency, and pulse duration.

2. Experimental procedures

Fig. 1 (a) presents the schematic of the femtosecond laser setup utilized in the study. The laser operates at a wavelength of 1030nm, featuring a 40 μ m spot diameter and pulse repetition rates ranging from 1 kHz to 1 MHz. It delivers a maximum average output of 40W. The pulse duration spans from 263fs to 5000fs, targeting a scanning region of 4mm $\times$ 4mm with a 0.06mm inter-line distance.



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Fig. 1. (a) Schematic diagram of the sapphire/Invar alloy femtosecond laser welding equipment; (b) Diagram of sample size; (c) Diagram of laser scanning path.

The specimens employed commercially sourced sapphire (Al_2O_3 , Hefei Yuanjing Crystal Co. Ltd) and Invar alloys (Fe-36Ni, Qisheng Sage Material Co. Ltd), measuring $7\text{mm} \times 7\text{mm} \times 3\text{mm}$ and $10\text{mm} \times 10\text{mm} \times 3\text{mm}$ (As shown in Fig. 1(b)), respectively.

The surface roughness of specimens critically determines the optical contact bonding performance. Therefore, nanoscale polishing is applied to the welding surfaces of both sapphire and Invar alloy, followed by ultrasonic cleaning and storage in a dust-free container to ensure the absence of scratches, particles, or contaminants on the bonding surfaces. During the welding process, assembled sapphire and Invar components are securely clamped using fixtures to achieve intimate surface contact, as illustrated in [Fig. 1\(a\)](#).

This method prevented the formed plasma and heat dissipation from the gap during femtosecond laser irradiation, which is essential for the high-strength joining of sapphire and Invar alloy using femtosecond laser technology. The parameters governing the femtosecond laser joining process for sapphire to Invar alloy integration are summarized in [Table 1](#).

Table 1. Femtosecond laser experimental parameters.

Parameter	Value	Unit
Power	3–15	W
Frequency	200–1000	kHz
Pulse duration	263–5000	fs
Scanning speed	50	mm/s
Defocus	–0.5	mm
Scan gap	60	μm
Scanning times	3	

Various laser processing parameters correlate with distinct laser energy inputs. For comparative analysis, the equation for peak power density, incorporating the three variables of laser power, pulse duration, and pulse frequency, is standardized by the

following equations.

$$E = \frac{P}{f} \quad (1)$$

$$P_{Peak} = \frac{4P}{\pi d^2 f t_p} \quad (2)$$

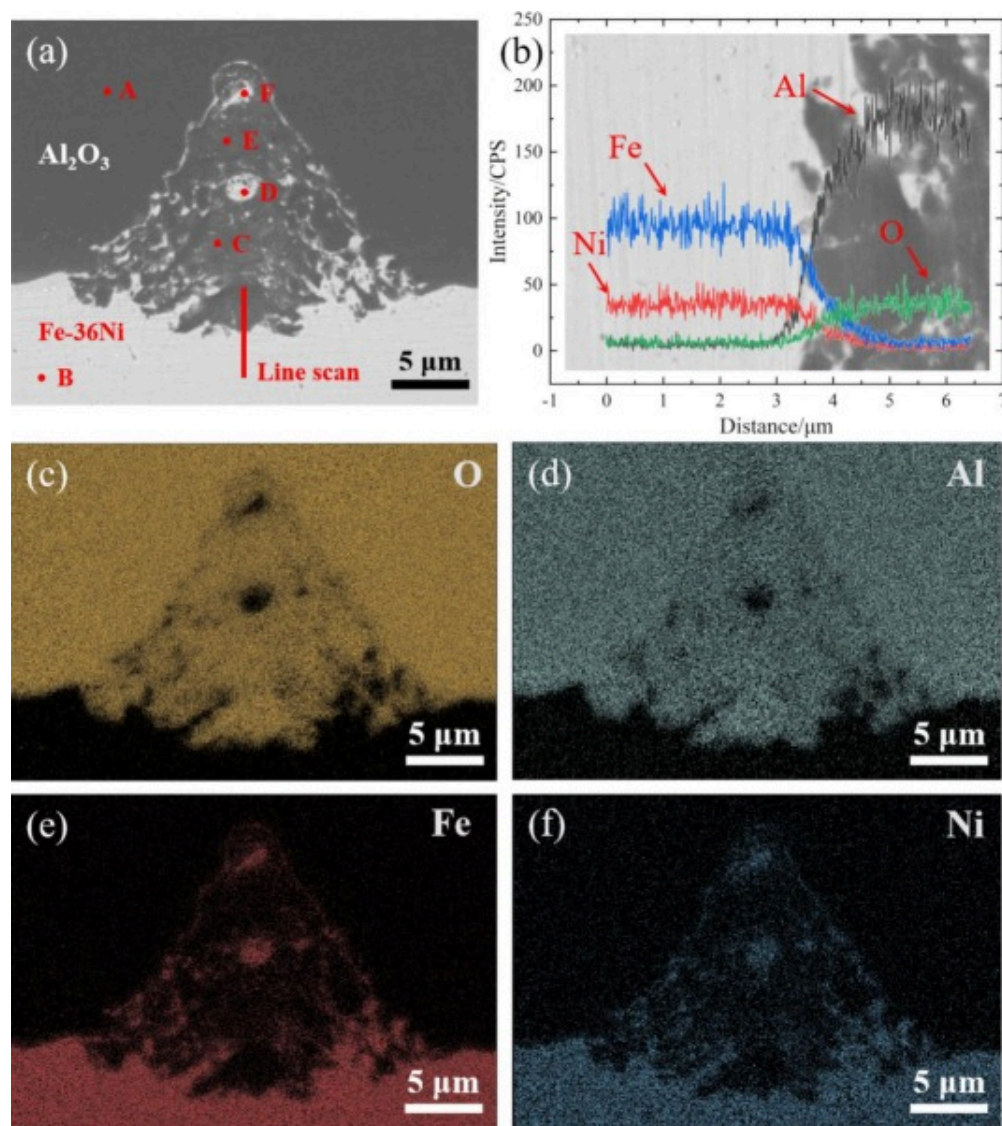
Where E is the single-pulse energy, P is the average laser power, f is the pulse frequency, P_{Peak} is the peak power density, d is the spot diameter, and t_p is the pulse duration. From eqs. (1), (2), it can be seen that the single-pulse energy is related to the average laser power and pulse frequency, and the single-pulse energy increases with the increase of the average laser power and the decrease of the pulse frequency, respectively. The peak power density increases with the increase of laser average power, the decrease of pulse frequency, and the decrease of pulse duration, respectively.

Completing the sapphire and Invar alloy joints was followed by analyzing the interface's macroscopic morphology utilizing a laser scanning confocal microscope (LSCM, Olympus OLS4100). Subsequently, the joint was cut by a diamond wire cutter, and its cross-section was meticulously polished to achieve a mirror finish. The microstructural characteristics of the joints were extensively examined through scanning electron microscopy (SEM, Quanta 650) and transmission electron microscopy (TEM, JEM-2100), both complemented by an energy dispersive spectroscopy (EDS, AMETEK) apparatus. The mechanical integrity of the joints was quantitatively evaluated via shear strength test, utilizing a universal mechanical testing machine (MTS-300) with a dedicated fixture designed to exert pressure at a uniform displacement rate of 0.5 mm/min. At least three specimens were tested under each condition to ascertain the mean strength value. Post-shear testing, the fracture surfaces on the Invar alloy and sapphire sides were subjected to SEM analysis.

3. Results and discussion

3.1. Microstructure analysis of a typical joint

Femtosecond laser welding of sapphire and Invar alloy produced typical joint morphologies at an average laser power of 7W, a frequency of 200kHz, and a pulse duration of 800fs (with a single-pulse energy of 35μJ and a peak power density of $3.48 \times 10^{12} \text{W/cm}^2$). Fig. 2(a) demonstrates that the sapphire/Invar joint displays an excellent bonding without structural defects, including voids or microcracks. Contrary to the typical morphology of conventional laser welded metal-to-metal joint, the sapphire/Invar alloy femtosecond laser-welded joint demonstrates an inverted molten pool shape. This inversion, characterized by the upward extension of the molten pool towards the sapphire, results from the thermally induced expansion of the plasma in an upward direction around the interface.



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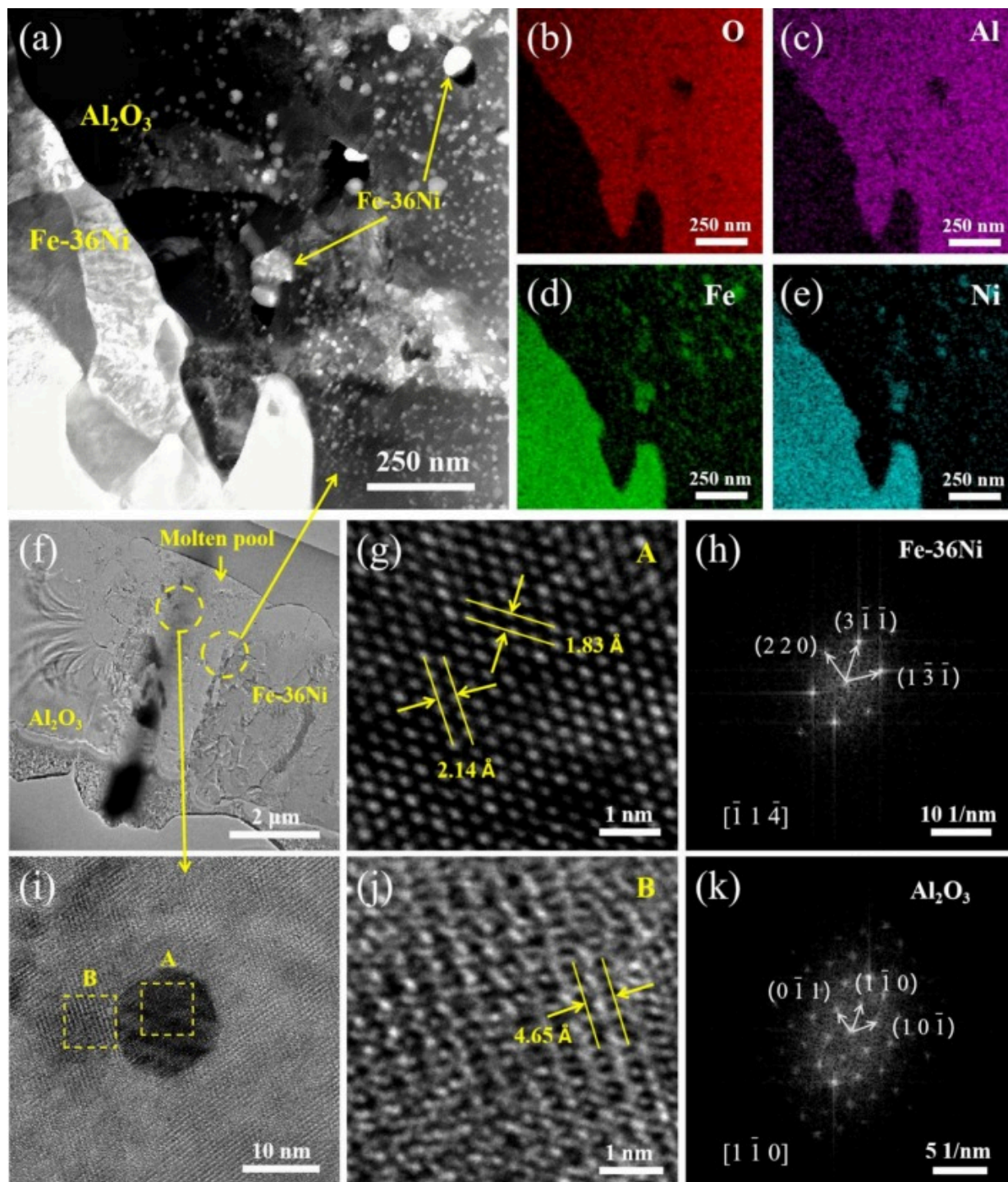
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Fig. 2. (a) Typical molten pool morphology; (b) EDS line scan results marked by the red line at the bottom of the molten pool; (c)-(f) EDS mapping results of the typical molten

pool. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

[Fig. 2](#) and [Table 1](#) illustrate the elemental distribution within the welding joint. [Fig. 2\(c\)-\(f\)](#) demonstrate that the O and Al elements do not infiltrate the Invar alloy, while Fe and Ni show significant diffusion into the molten pool. [Fig. 2\(b\)](#) demonstrates that there are no notable metallurgical reactions at the interface of sapphire and Invar alloy. This outcome is due to the inadequate metallurgical compatibility between Al_2O_3 and Fe-36Ni. The data presented in [Table 1](#) and [Fig. 2](#) indicate that the possible phases within the molten pool primarily consist of Al_2O_3 and Fe-36Ni.

To further corroborate the phase composition within the welding joint, analyses of TEM bright-field and dark-field images were conducted on the molten pool, as depicted in [Fig. 3](#). As observed from [Fig. 2\(a\)-\(f\)](#), the majority of the interior of the joint is comprised of phase B. Within the joint, the phase B matrix contains spherical phase A of different sizes. The evidence presented in the TEM high-resolution Fourier diagrams in [Fig. 3\(f\)-\(k\)](#) identifies phase A as the Fe-36Ni alloy and phase B as Al_2O_3 , supporting the hypotheses outlined in [Table 2](#) concerning the phase composition within the joint interior. The examination of the high-resolution Fourier diagrams obtained from TEM analysis of the joint interior confirms that the interaction enabled by the femtosecond laser does not cause metallurgical reactions between sapphire and the Invar alloy in the joint, nor does it lead to the emergence of new phases. The formation of the joints can be attributed to the physical mixture of sapphire and Invar alloys.



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Fig. 3. TEM analysis of the sapphire/Invar joint interface: (a)–(e) dark field images of the sapphire/Invar joint region and EDS; (f)–(k) high-resolution maps of the interior of the sapphire/Invar joint and Fourier plots.

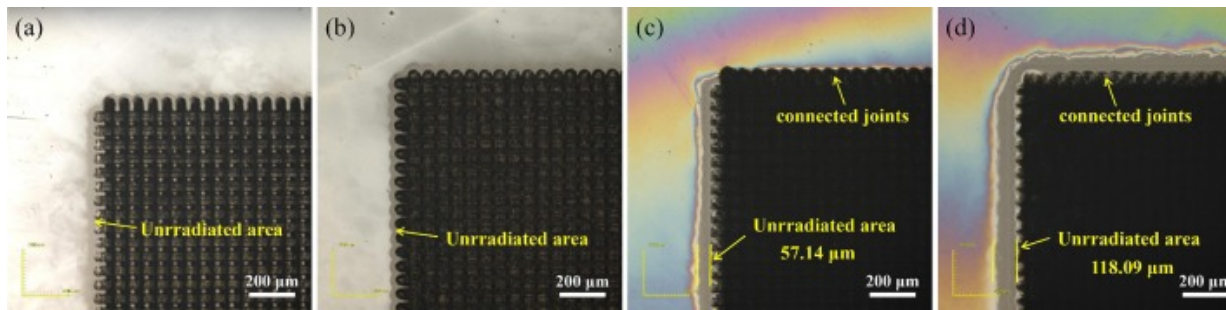
Table 2. EDS results at Points A–F marked in joint shown in [Fig. 2 \(a\)](#).

Elements (at.%)	O	Al	Fe	Ni	Possible phases
Point A	60.82	39.08	0.07	0.03	Al ₂ O ₃
Point B	6.68	1.16	59.47	32.68	Fe-36Ni
Point C	58.01	37.50	3.13	1.37	Al ₂ O ₃
Point D	24.97	16.59	36.36	22.07	Al ₂ O ₃ , Fe-36Ni
Point E	58.79	37.56	2.50	1.15	Al ₂ O ₃
Point F	39.48	28.50	19.78	12.24	Al ₂ O ₃ , Fe-36Ni

3.2. Effect of average power on morphology and microstructure of Sapphire/Invar joint

[Fig. 4](#) illustrates the macroscopic morphology of sapphire/Invar alloy joints at varying average laser powers. [Fig. 4 \(a\)–\(d\)](#) demonstrate that the melting range in the laser-unirradiated zone increases progressively with higher average power, which is due to the rise in single-pulse energy and peak power density. At an average power of 11 W, the melting width in the laser-unirradiated zone is approximately equal to the scan line spacing. At an average power of 15 W, the melting width of the laser unirradiated region is 118.09 μm, which is approximately twice the scan line spacing, and it can be seen that the internal adjacent weld paths completely overlap. The overlap markedly increases the

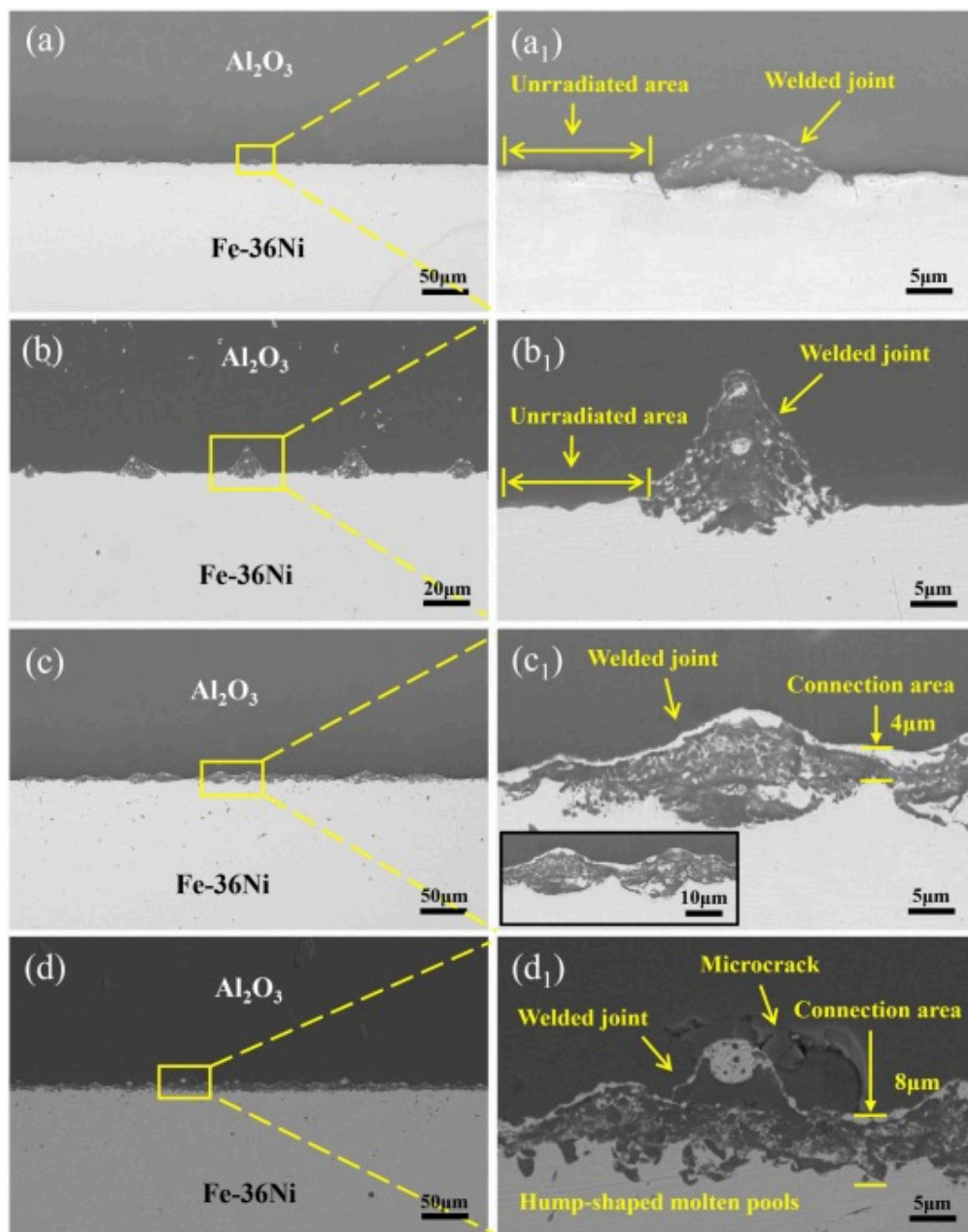
thickness of the molten region in the laser-unirradiated zone, as demonstrated in Fig. 5 (c_1) and (d_1).



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Fig. 4. Macroscopic morphology of the sapphire/Invar alloy joints welded with different average powers: (a) 3W; (b) 7W; (c) 11W; (d) 15W.



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Fig. 5. Microstructure and morphology of the sapphire/Invar alloy joint welded with different average powers: (a)-(a₁) 3W; (b)-(b₁) 7W; (c)-(c₁) 11W; (d)-(d₁) 15W.

The microstructure of the sapphire/Invar alloy femtosecond laser welded joints, illustrated in Fig. 5, shows a relationship between the average power and the dimensions of the joints, specifically their width and depth. An initial increase in joint dimensions occurs with rising power levels, followed by a decrease. Fig. 5 (a) demonstrates that at an average power of 3W (single pulse energy of 15μJ and a peak power density of $1.49 \times 10^{12} \text{W/cm}^2$), the dimensions of the joint are 17.5μm in width and 5.625μm in depth, attributable to the decreased single pulse energy and peak power density. Under these conditions, the joint primarily consists of sapphire, with a negligible presence of Invar alloy.

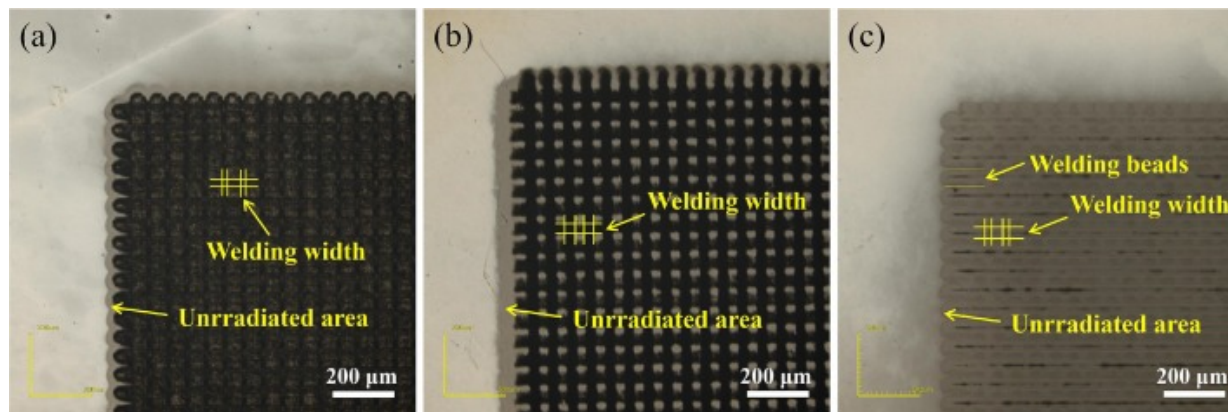
As shown in Fig. 5 (b)-(b₁), when the average power was increased to 7W (single pulse energy of 35μJ and peak power density of $3.48 \times 10^{12} \text{W/cm}^2$), the transverse and longitudinal dimensions of the molten pool were increased due to the increased single pulse energy and peak power density. The degree of mixing of sapphire and Invar alloy inside the joint also increased, with a large amount of Invar alloy incorporated inside.

Fig. 5 (c)-(c₁) demonstrate that an increase in average power to 11W (single pulse energy of 55μJ and a peak power density of $5.47 \times 10^{12} \text{W/cm}^2$), results in a further increase in both single pulse energy and peak power density. Additionally, the transverse dimensions of the joints expand, leading to a more uniform mixing of the internal sapphire with the Invar alloy. The figure illustrates that the formation of hump-shaped joints within the focus range and the connection phenomenon observed in the laser-unirradiated area between adjacent joints can be attributed to two potential reasons. Firstly, an increase in laser power and energy, along with an improved heat transfer range, may lead to the melting of the laser-unirradiated area. Alternatively, the internal plasma may absorb laser energy, leading to energy deposition and subsequent plasma expansion on both sides, thus increasing the melting range.

Fig. 5(d)-(d₁) demonstrate that an increase in average power to 15W (single pulse energy of 75μJ and a peak power density of $7.46 \times 10^{12} \text{W/cm}^2$) resulted in the maximum values for both alone pulse energy and peak power density. The figure illustrates a hump-shaped joint presents in the focal range, with an increased thickness of the melt in the unirradiated area, accompanied with complete melting of the entire interface. Notably, excessive energy input and resultant thermal stress lead to the development of microcracks in the sapphire adjacent to the joint, making it vulnerable to stress concentration under applied stress. This is harmful to the mechanical properties of the joint.

3.3. Effect of repetition frequency on morphology and microstructure of Sapphire/Invar joint

Fig. 6 illustrates the macroscopic morphology of the sapphire/Invar joints at different femtosecond laser repetition frequencies. Fig. 6 (a)-(c) demonstrate a reduction in the ablation degree at the macroscopic joint interface between sapphire and Invar alloy with an increase in the repetition frequency of the femtosecond laser. This results from the decreased peak power density caused by the competitive relationship between repetition frequency and single pulse energy under the same laser power, which lessens the ablation effect on the sapphire/Invar alloy.

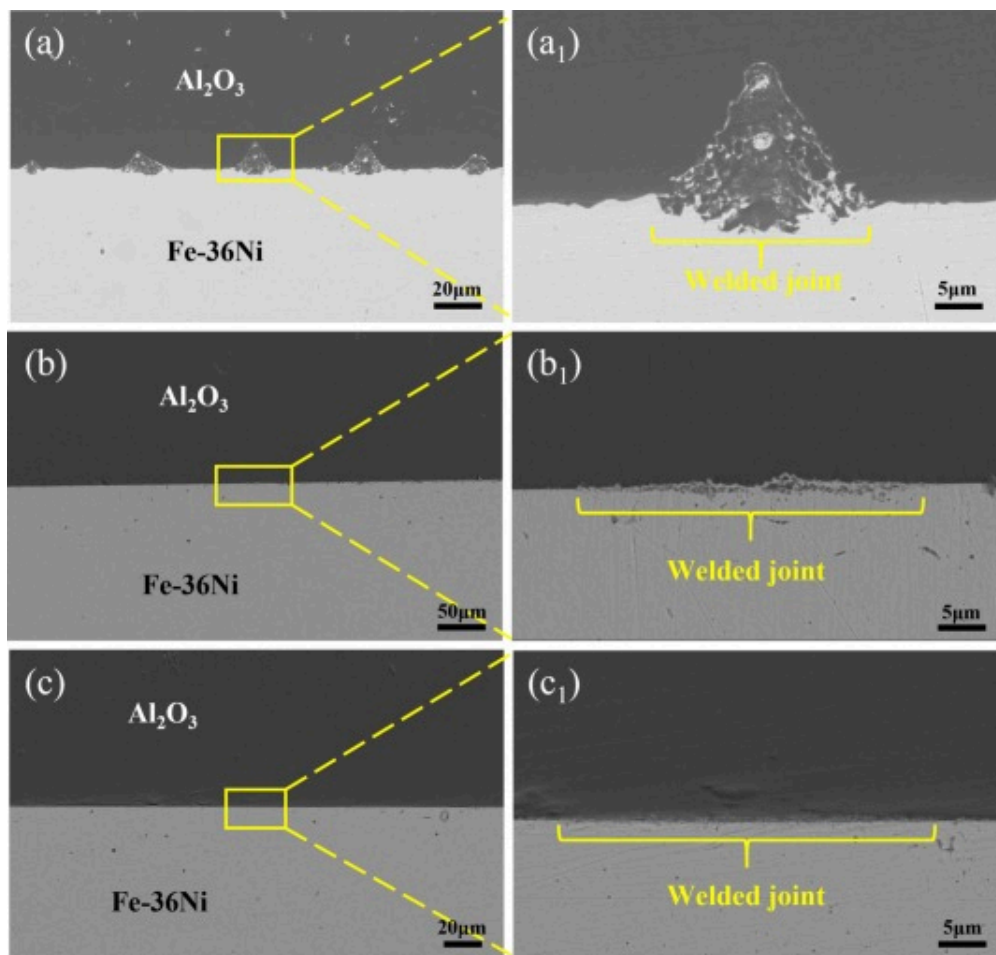


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Fig. 6. Macroscopic morphology of the sapphire/Invar alloy joints at different laser frequencies: (a) 200kHz; (b) 600kHz; (c) 1000kHz.

The microscopic morphology of the sapphire/Invar alloy joint at different femtosecond laser frequencies is shown in Fig. 7. Analysis indicates a decrease in both single pulse energy and peak power with escalating laser frequency, consequently diminishing the ablation efficiency on the sapphire/Invar alloy per pulse. At a frequency of 200kHz (a single pulse energy of $35\mu\text{J}$ and a peak power density of $3.48 \times 10^{12} \text{W/cm}^2$), a large joint size and a good mix of sapphire and Invar alloy within the joint are observed, as illustrated in Fig. 7 (a₁). When the frequency reaches 600kHz (single pulse energy of $11.67\mu\text{J}$, peak power density of $1.16 \times 10^{12} \text{W/cm}^2$), the reduction in single pulse energy and peak power density decreases the longitudinal depth of the joint, with sapphire particles within the joint, as shown in Fig. 7 (b₁). As the pulse frequency escalates to 1000kHz, both the single pulse energy and peak power density further decrease (single pulse energy of $7\mu\text{J}$, peak power density of $0.696 \times 10^{12} \text{W/cm}^2$). The reduction in peak power density leads to weakened nonlinear absorption, which decreases the energy deposition of the laser at the interface. Consequently, plasma formation is constrained to lateral eruptions at the interface rather than in an upward direction, leading to an expansion in transverse width and a reduction in longitudinal depth. Therefore, it is suggested that a low frequency should be employed for a reliable bonding of sapphire and Invar alloys.



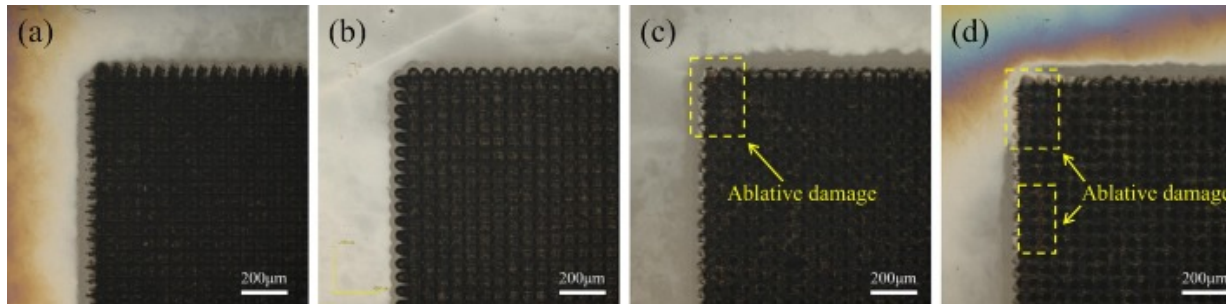
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Fig. 7. Microscopic morphology of the sapphire/Invar alloy joints at different laser frequencies: (a)-(a₁) 200kHz; (b)-(b₁) 600kHz; (c)-(c₁) 1000kHz.

3.4. Effect of pulse duration on morphology and microstructure of Sapphire/Invar joint

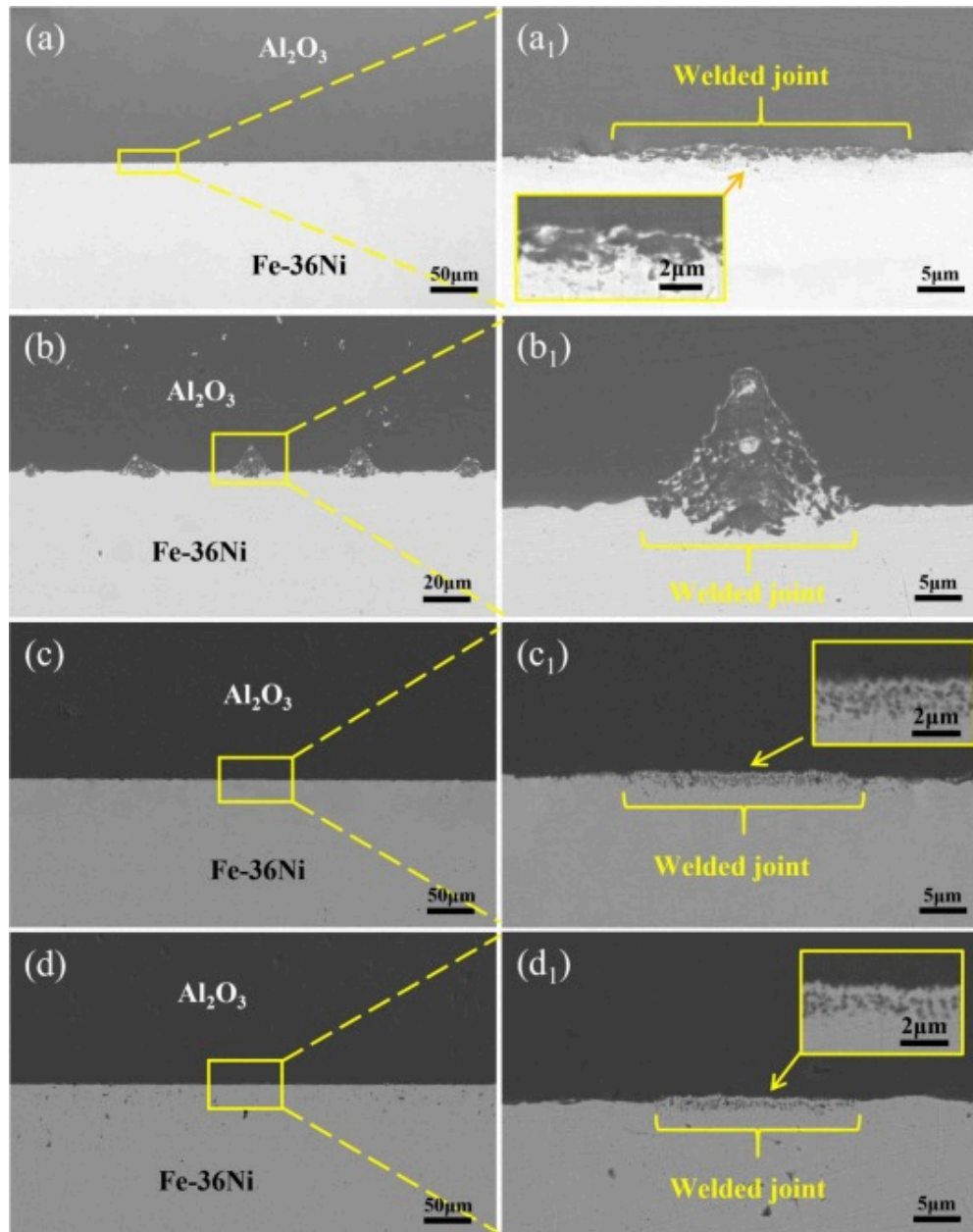
As shown in Fig. 8, Fig. 9, the study presents the macroscopic and microscopic morphologies of sapphire/Invar alloy joints subjected to varying pulse durations (maintaining a constant single pulse energy of $35\mu\text{J}$) at a pulse frequency of 200kHz and an average laser power of 7W . Ablation damage is observed at the sapphire and Invar interfaces at 3000fs and 5000fs , attributed to an increased thermal effect caused by the pulse durations larger than the time scale of the interaction between the electron and phonons.



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Fig. 8. Macroscopic morphology of sapphire/Invar alloy joints at different pulse durations: (a) 263fs ; (b) 800fs ; (c) 3000fs ; (d) 5000fs .



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Fig. 9. Microscopic morphology of sapphire/Invar joints at different pulse durations: (a)-(a₁) 263fs; (b)-(b₁) 800fs; (c)-(c₁) 3000fs; (d)-(d₁) 5000fs.

As illustrated in Fig. 9, there is a discernible variation in joint size that correlates with alterations in pulse duration. The analysis demonstrates that the joints produced at pulse durations of 263fs, 3000fs, and 5000fs (corresponding to peak power densities of $11.58 \times 10^{12} \text{W/cm}^2$, $0.93 \times 10^{12} \text{W/cm}^2$, and $0.5568 \times 10^{12} \text{W/cm}^2$, respectively) display similar dimensions and flat geometries. This is also due to a reduction in peak power density, which leads to a weakening of nonlinear absorption, reducing the energy deposition of the pulse laser on the surface of the material. However, unlike 3000fs and 5000fs, when the pulse duration was 263fs, larger internal sapphire particles were observed within the joints. This phenomenon is due to the fact that the 263fs pulse duration has a higher peak power density than that corresponding to the 3000fs and 5000fs. Despite this, the interface between the joint and the sapphire substrate in the configurations utilizing pulse durations of 3000fs and 5000fs is characterized by a jagged morphology, which, to some extent, enlarges the contact interface between the joint and the substrate, as demonstrated in Fig. 9(c₁) and (d₁). Conversely, the interface at a pulse duration of 263fs is markedly smoother, as illustrated in Fig. 9(a₁). Increasing the pulse width from 263fs (peak power density of $11.58 \times 10^{12} \text{W/cm}^2$) to 800fs (peak power density of $3.48 \times 10^{12} \text{W/cm}^2$) results in a decrease in peak power density, while the joint size increases. This phenomenon occurs probably because the 263fs pulse duration is insufficient for effective material interaction with the joint. Consequently, the chosen pulse duration must guarantee a peak power density adequate for significant nonlinear absorption, while also allowing for sufficient pulse interaction time.

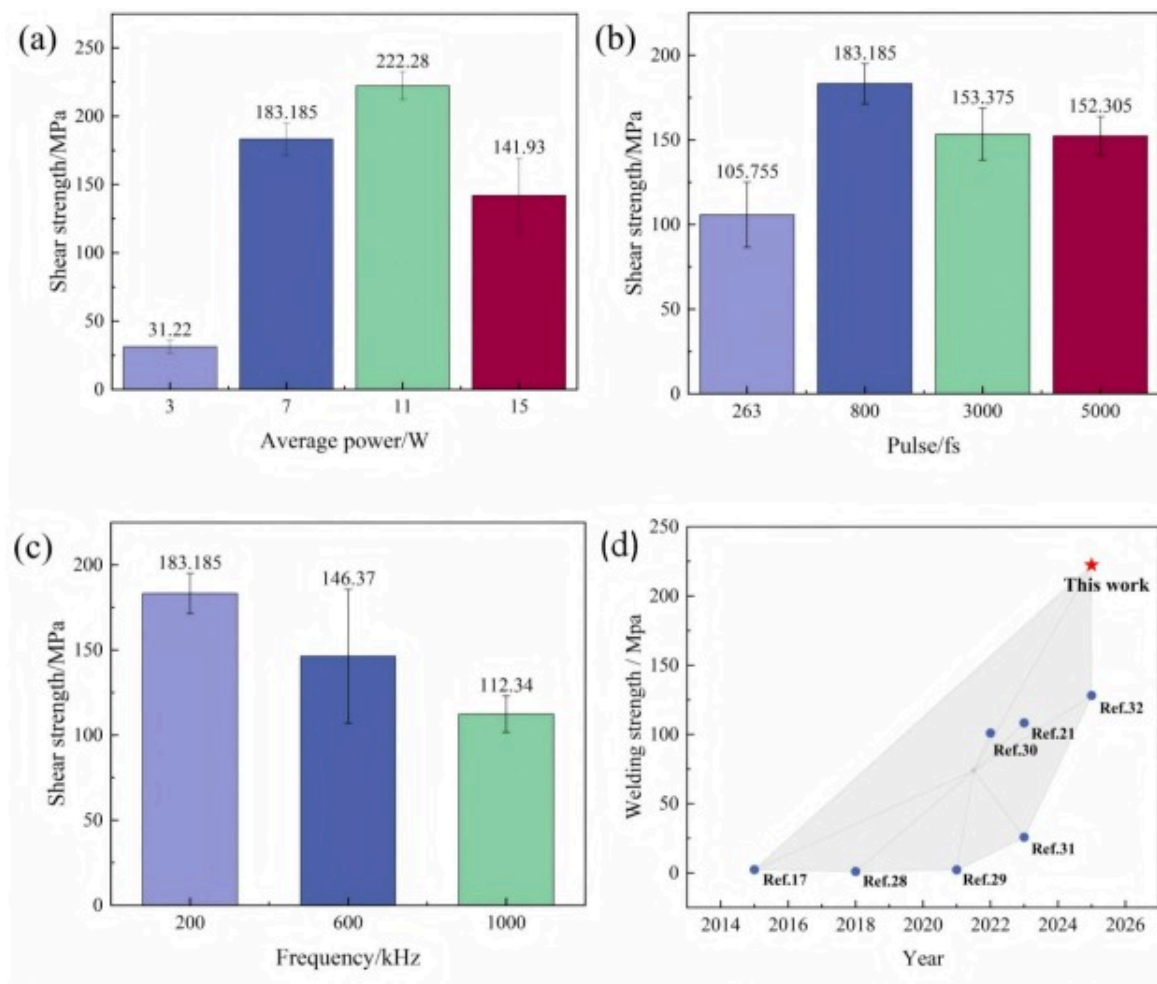
In conclusion, the three experiments above indicate that when the peak power density is below $1.16 \times 10^{12} \text{W/cm}^2$, the interfacial materials may not achieve sufficient melting, resulting in small longitudinal dimensions of the formed joint. Furthermore, when the pulse width is excessively small, despite a high peak power density, the duration of the pulse interacting with the materials is insufficient, also perhaps leading to inadequate melting of the interface materials. When the peak power density exceeds

$7.46 \times 10^{12} \text{ W/cm}^2$, microcracks occur in the joint. Optimal joint formation with a distinct embedded structure is attainable only when the peak power density falls within the range of $1.49 \times 10^{12} \text{ W/cm}^2$ to $5.47 \times 10^{12} \text{ W/cm}^2$.

4. Mechanical property

4.1. Relationship between shear strength and joint microstructure

[Fig. 10](#) shows the shear strength of the sapphire/Invar alloy joint at different parameters. [Fig. 10](#) (a) illustrates the shear strength of sapphire/Invar alloy joints at different average laser powers. The shear strength initially increases and subsequently decreases as power increases, where the maximum shear strength of the joint is 222.28 MPa. The reduction in shear strength can be linked to elevated thermal stresses in the sapphire resulting from high laser energy input, which induces microcracks and subsequently diminishes the shear strength of the joint, as illustrated in [Fig. 5\(d₁\)](#).



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Fig. 10. Shear strength of sapphire/Invar joints: (a) 3W–15W; (b) 263fs–5000fs; (c) 200kHz–1000kHz; (d) Strength comparison of metal/transparent material welded by ultrafast laser via optical contact in recent years [[27], [28], [29], [30], [31]].

Fig. 10 (b) shows the shear strength of sapphire/Invar alloy joint at different pulse durations. A maximum shear strength of 183.15MPa was obtained at 800fs, which is

attributed to its good joint formation and the homogeneous mixing of the materials within the molten pool. The shear strength values of the joints at 3000fs and 5000fs are similar, with both exceeding the value at 263fs. The joints at 3000fs and 5000fs exhibit jagged sapphire/Invar alloy interfaces, resulting in a physically interlocked structure that enhances joint shear strength. This contrasts the relatively smooth morphology of the joint interface observed at 263fs, as illustrated in Fig. 9 (a₁), (c₁), and (d₁).

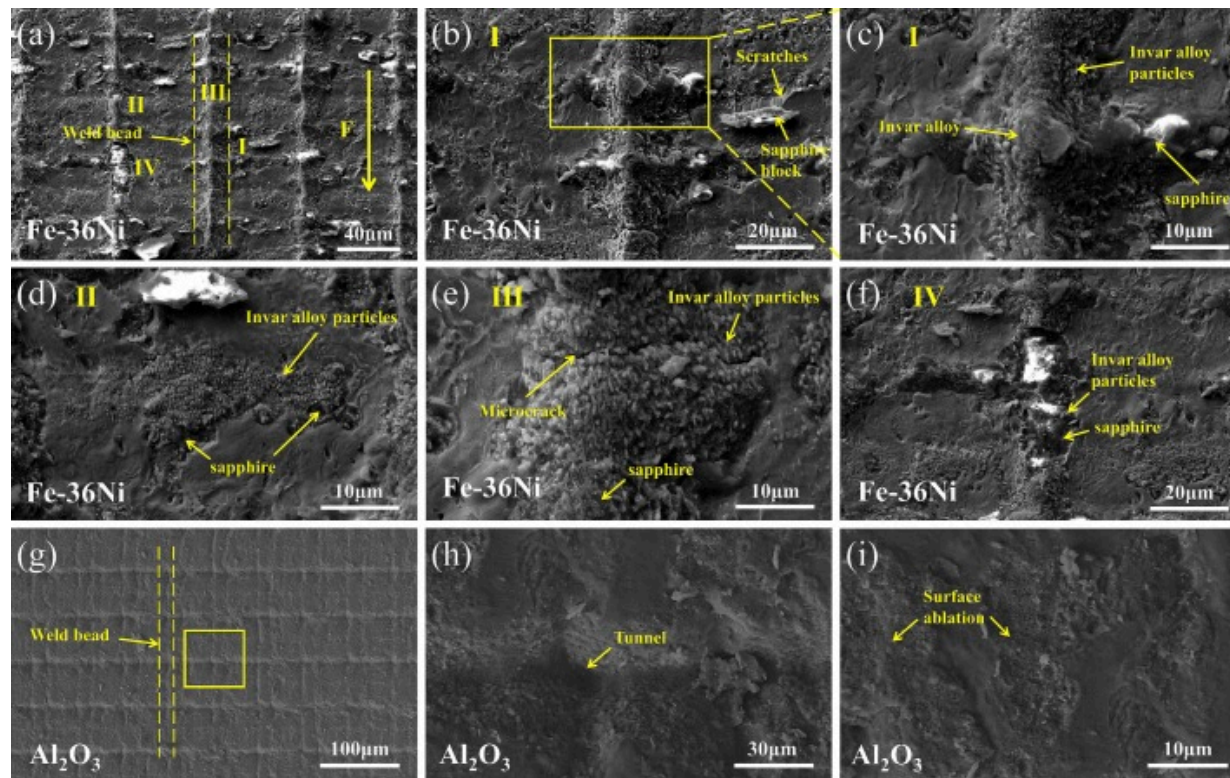
Fig. 10 (c) shows the shear strength of the sapphire/Invar alloy joint at different pulse frequencies. The joint shear strength decreases as frequency increases. This phenomenon occurs because a reduction of single pulse energy and peak power density corresponding to higher frequencies leads to a smaller molten pool and diminished interlocking effect of the joint, ultimately reducing the joint shear strength.

Fig. 10(d) presents the welding strength of ultrafast laser-welded joints for metal/brittle materials achieved through optical contact in recent years. The maximum shear strength obtained in this study significantly surpasses that reported in other research of the same category.

4.2. Typical fracture micro-morphology of Sapphire/Invar joint

Fig. 11 depicts a typical fracture morphology of a sapphire/Invar alloy joint achieved at an average laser power of 7W, pulse duration of 800fs, and pulse frequency of 200kHz. The fracture location on the Invar alloy side exhibits convex joints, as illustrated in Fig. 11 (a-f). The fracture surface on the side of sapphire was left with concave pits (Fig. 11 g-i), which, together with the convex joints on the Invar fracture surface, constituted a mechanical interlocking structure. As a result of plastic deformation caused by shear forces, visible scratches and damaged joint domain appear at the weld fracture interface with the Invar alloy matrix. Additionally, Fig. 11 (d) and (i) illustrate the regions within the laser-unirradiated areas at the sapphire/Invar alloy interface, demonstrating varying levels of ablation and deposition of Invar alloy particles on the ablated surfaces. This phenomenon occurs when laser-induced periodic surface structures (LIPSS) are formed

on the material surface as the laser energy slightly exceeds the material's ablation threshold, creating particulate structures at elevated energies [32]. Therefore, the shear strength of the sapphire/Invar alloy joint arises from two main aspects: the mechanical interlock of convex joints with concave pits and the bonding between joints and the substrate.



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Fig. 11. Typical morphology of joint fracture surfaces: (a) fracture morphology of Invar alloy; (b)-(c) convex joint at the Invar fracture interface; (d) metal ablation morphology in the laser-unirradiated region; (e) mixing morphology of Invar alloy particles and sapphire; (f) fracture morphology of the joint with the Invar substrate; (g) fracture

morphology of sapphire; (h) concave pit at the sapphire fracture interface; (i) ablation morphology of sapphire in the laser-unirradiated region.

5. Conclusions

In conclusion, the strong bond between sapphire and Invar alloy was achieved utilizing femtosecond laser welding, resulting in distinct joint features. The sapphire/Invar alloy joint is well-built and devoid of flaws, including voids and fractures. The joint exhibits a mosaic structure formed from a sapphire and Invar alloy mixture devoid of any chemical metallurgical reaction. The shear strength of the sapphire/Invar alloy joint reaches 222.28MPa, about 50% higher than the shear strength of the joints previously examined by our research. The shear strength of the joints primarily results from three factors: first, the mechanical interlocking effect created by the mosaic structure; second, the bond between the joints and the substrate; and third, the bonding in the laser-unirradiated region.

For femtosecond laser welding of sapphire/Invar alloys, two conditions are required to obtain joints with high-strength mosaic structures: i) the pulse duration needs to exceed the minimum welding threshold of 800fs, as shorter pulse durations do not allow sufficient interaction time between individual pulses and the materials. ii) the peak power density needs to be in a range from $1.49 \times 10^{12} \text{W/cm}^2$ to $5.47 \times 10^{12} \text{W/cm}^2$.

CRedit authorship contribution statement

Rui Pan: Writing – original draft, Conceptualization. **Fuqiang Ye:** Writing – original draft, Methodology, Investigation. **Yinghao Feng:** Validation, Data curation. **Pei Chen:** Validation, Supervision. **Jin Yang:** Resources, Methodology. **Peng Li:** Visualization, Resources. **Shujun Chen:** Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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